

ALBERT JOHN C. DIAZ

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PROFESSIONAL SUMMARY

Computer Engineering graduate with hands-on experience in embedded systems, IoT firmware development, and full-stack software engineering. Proven ability to design and integrate hardware-software solutions, utilizing microcontrollers, IoT protocols, and machine learning to build end-to-end monitoring and recommendation systems. Proficient in Python, C/C++, JavaScript (React, Node.js), and PostgreSQL, with a solid foundation in network fundamentals (CCNA). Civil Service Professional-eligible, recognized for strong technical documentation, cross-functional team leadership, and a highly adaptable, detail-oriented approach to complex problem-solving. Open to relocation and nationwide opportunities.

EDUCATION

Bachelor of Science in Computer Engineering

June 2026

University of Science and Technology of Southern Philippines (USTP)

Dean's Lister (2022–2023)

WORK EXPERIENCE

Firmware & Software Engineering Intern

January 2026 – March 2026

Columbia Computer Center Inc. – CDO Branch

- Developed and maintained full-stack web applications using React, TypeScript, Node.js, and PostgreSQL, applying iterative delivery practices aligned with Agile workflows.
- Performed hardware diagnostics and computer repair tasks, gaining hands-on exposure to digital circuit troubleshooting and system integration.
- Contributed to job order management processes, coordinating between technical and administrative teams to ensure timely service delivery.
- Documented technical procedures and system configurations to support operational compliance and knowledge transfer.

PROJECTS

TANIM: IoT Soil Health Monitoring & Crop Recommendation System

May 2025 – April 2026

ESP32 microcontrollers · Modbus-RTU · RS485/UART · LoRa · GPS · IoT sensors · Machine Learning (Light GBM) · Embedded C · Agile iterations · Solar Power Management

- Designed and implemented embedded firmware for ESP32 microcontrollers to interface with a 7-in-1 soil nutrient sensor over Modbus-RTU, routing communication through an RS485-to-TTL adapter via UART handling register-level read commands, response parsing, and fault detection for robust real-time soil data acquisition.
- Architected a long-range wireless communication layer using LoRa for transmission between field nodes and a gateway, enabling reliable IoT data pipelines across remote agricultural deployments with limited cellular coverage.
- Integrated a GPS module for geo-tagged soil readings, allowing per-location tracking of soil health data across multiple farm plots and enabling location-aware crop recommendations.
- Implemented solar power management logic in firmware to monitor battery state, regulate sensor polling intervals under low-charge conditions, and prioritize critical transmissions ensuring continuous field operation without manual intervention.

Job Order Management System

February 2026 – March 2026

React · TypeScript · Node.js · Express.js · PostgreSQL · Tailwind CSS · Git

- Built a production-grade web application with structured state management, role-based access, and real-time status monitoring, demonstrating systematic software development practices.
- Implemented end-to-end testing and validation procedures, ensuring system reliability and traceability of repair and service workflows.

- Maintained version control using Git with documented commit history, supporting collaborative development and code review processes.

IoT-Based Forest Monitoring System

March 2025 – June 2025

ESP32 · IoT · Embedded Firmware · Sound Sensor · ESP32-CAM · HTTP/Wi-Fi

- Designed and implemented event-driven, state-machine firmware on ESP32 microcontrollers to continuously poll an audio sensor and camera module, classifying environmental sounds in real time to detect unauthorized logging and poaching activity.
- Architected firmware states (IDLE → SENSING → DETECTION → ALERT → RESET) to manage concurrent sensor polling, threshold-based event triggering, and alert transmission over Wi-Fi using HTTP, ensuring reliable operation in low-power field deployments.
- Integrated ESP32-CAM for image capture on detection events, coordinating data flow between peripheral modules via GPIO and hardware interfaces.

MART: Market of ART

December 2024

Kotlin · Android SDK · Mobile Application

- Led the team in designing and developing a cross-functional e-commerce mobile application in Kotlin, demonstrating proficiency in structured software architecture and mobile UI development.

CERTIFICATIONS & TRAINING

- Civil Service Eligibility – Professional Level, Civil Service Commission, Philippines (Passed: May 2026)
- CISCO CCNA: Enterprise Networking, Security, and Automation
- CISCO CCNA: Switching, Routing, and Wireless Essentials
- CISCO: Introduction to Cybersecurity
- Educational Exposure: Profood International Corp., National Grid Corporation of the Philippines, Globe, OnSemi

TECHNICAL SKILLS

Embedded & Hardware: ESP32, Arduino, ARM-based microcontrollers, IoT sensors, I2C/SPI/UART protocols, Electronic diagnostics, CCNA Networking, AutoCAD

PCB Design: Autodesk Eagle, EasyEDA, KiCAD

Languages & Frameworks : JavaScript/TypeScript, React, CSS, HTML, Embedded C, C/C++, Python, Java, Kotlin, LaTeX, MATLAB, Assembly (MIPS), VHDL / Verilog, MATLAB

Databases & Architecture: PostgreSQL, MySQL, Firebase, Git/GitHub, VSCode, Antigravity, Microsoft Office, Google Workspace

Ai Tools: Claude, ChatGPT, Google Gemini, Deepseek, Lovable, Bolt, Perplexity, Zapier

Soft Skills: Technical documentation, problem-solving, cross-functional collaboration, team leadership, fast learner, adaptable, attention to detail,

LEADERSHIP & ACTIVITIES

- Year Representative, Institute of Computer Engineers of the Philippines Student Edition – USTP (2024–Present)
- Mobile Legends Representative (Captain), ICePEP.SE – USTP Regionals & National Events (2023–2025)

REFERENCES

Engr. Jodie Rey D. Fernandez - Center of Artificial Intelligence USTP | jodierey.fernandez@ustp.edu.ph

Engr. Rodesita S. Estenzo - Computer Engineering Chairperson USTP | rodesita.estenzo@ustp.edu.ph

Wellbien P. Cimagala - Columbia Computer Center Inc. – CDO Branch Manager | wellbien@yahoo.com